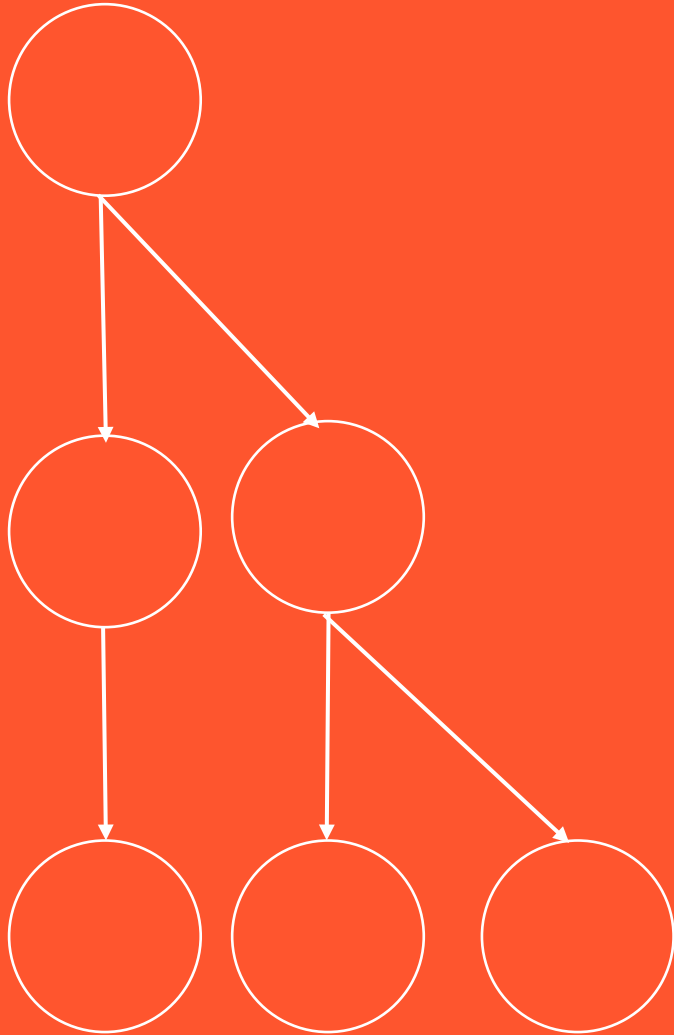


BCOG 100

Language

Lecture 1: Genus Homo
and the Evolution of Language



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Reading Check 11-1

Password: kanzi

Question: According to the reading, what is language?

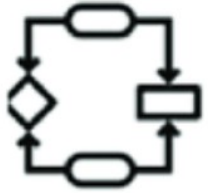


The Evolution of Homo Sapiens and Language



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

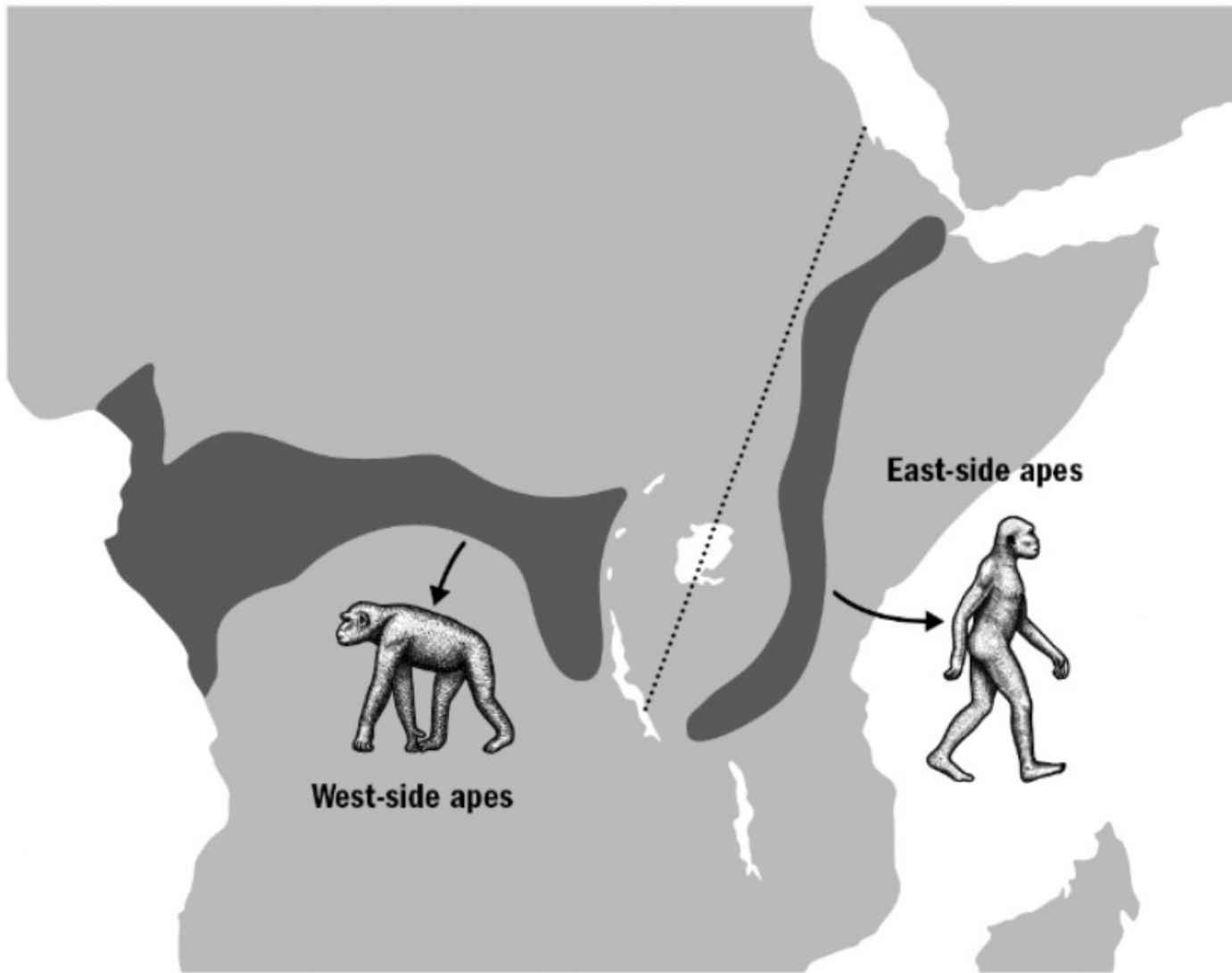
Last Time:

Social cognition and theory of mind in primates

Today:

This week: evolution of language and modern human minds

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From Apes to Early Homo

- Divergence from chimpanzees about seven million years ago
- Changing climate reshaped African ecosystems
- Early hominins adapted to new environments

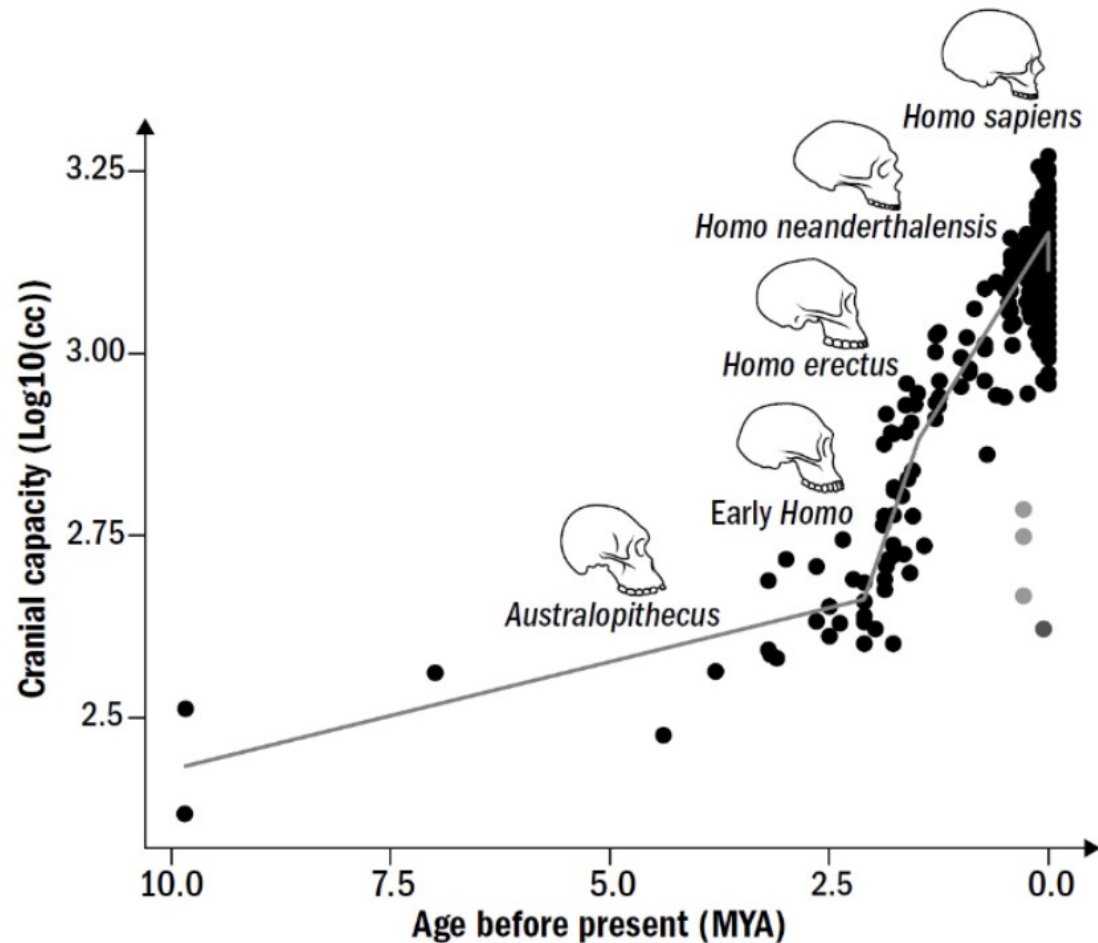
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The Savannah Brain: Homo erectus and New Adaptations?

- Homo erectus emerged about two million years ago
- Larger brains and bodies, new tools and behaviors
- Life on the savannah demanded coordination and learning

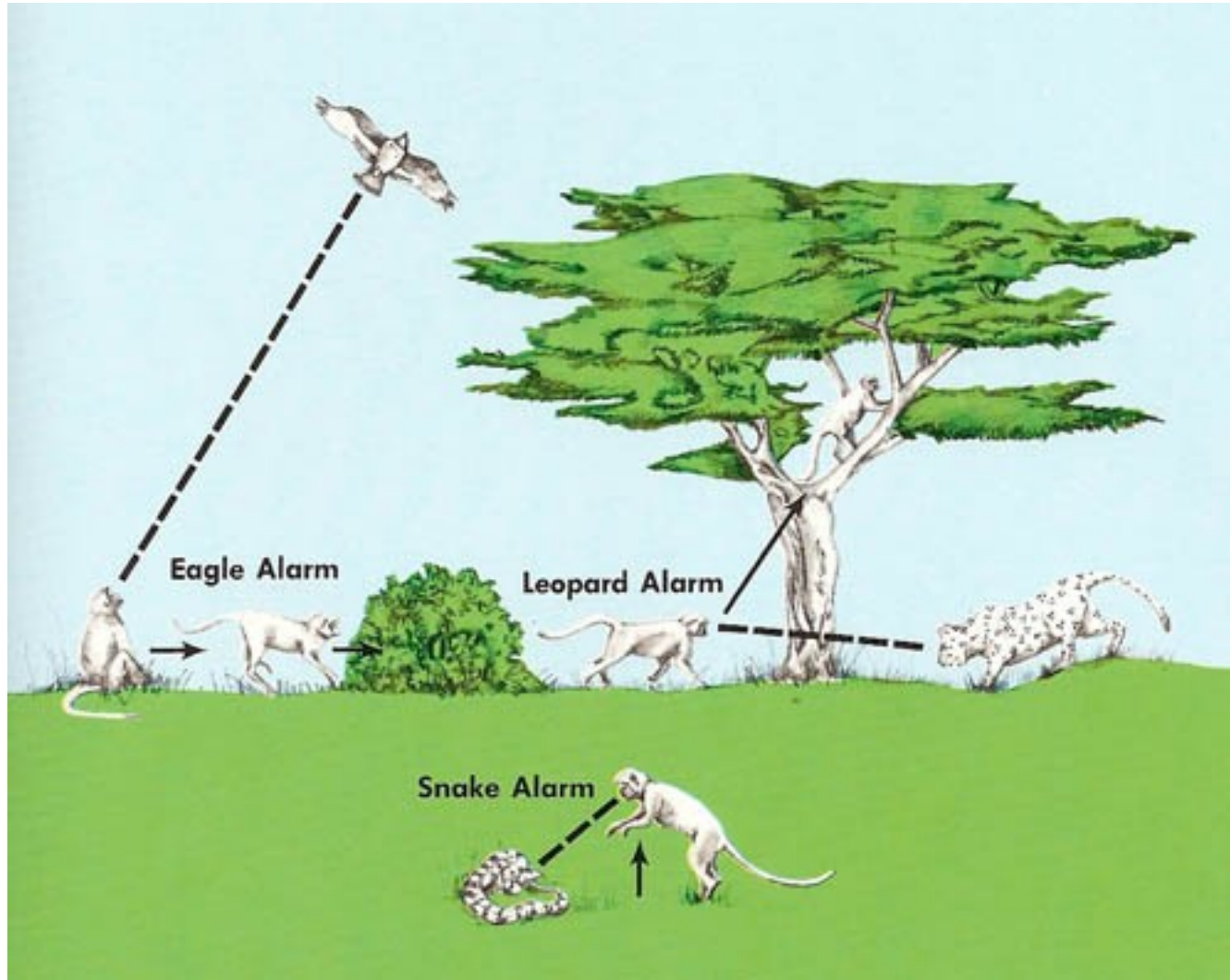
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So What Changed in the Mind?

- Early humans looked familiar but thought differently
- Brains grew, but new kinds of thinking mattered more
- Two key innovations: **language** and **symbolic thought**

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Communication Before Language

- Communication evolved long before humans
- Each lineage added new ways to share information
- Language built on this deep evolutionary foundation

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How Is Human Language Different? Hockett's Design Features

Feature	Brief Definition	Unique to Humans
Vocal-Auditory Channel	Language is typically spoken and heard.	
Broadcast Transmission & Directional Reception	Speech radiates in all directions but can be localized.	
Rapid Fading (Transitoriness)	Spoken signals fade quickly after being produced.	
Interchangeability	Speakers can both send and receive the same messages.	
Total Feedback	Speakers can hear and monitor their own speech.	
Specialization	Sounds are produced solely for communication.	
Semanticity	Specific signals have specific meanings.	
Arbitrariness	No natural link between form and meaning.	
Discreteness	Language uses distinct, separable units.	
Displacement	Ability to refer to things not present in time or space.	X
Productivity (Open-Endedness)	Ability to create and understand novel messages.	X
Traditional (Cultural) Transmission	Language is learned socially, not genetically fixed.	X
Duality of Patterning	Meaningless units combine into meaningful ones.	X
Prevarication	Ability to lie or produce false statements.	X
Reflexiveness	Ability to use language to talk about language.	X
Learnability	Speakers of one language can learn another.	X



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Animal Communication

Fixed meaning
("alarm = eagle")

Context-bound
(here and now)

Holistic signals

Emotional or instinctive

Primarily **instrumental**
(to get things)

Human Language

Arbitrary symbols
("word ↔ idea")

Displaced
(past, future, imagined)

Combinatorial grammar

Intentional and productive

Sometimes **declarative**
(to share information)

Symbols and Syntax as Key Innovations

- Arbitrary words freed meaning from emotion and context
- Grammar allowed endless combinations of ideas
- Together they made language generative and open-ended



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Gorilla Foundation / Box Cohn, File



Ape Communication in the Wild

- Apes use rich gestures, expressions, and calls
- Signals are flexible but rarely symbolic
- Communication stays tied to emotion and context

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Teaching Apes Human Language Systems

- Researchers have tried to teach apes human-like languages
- Famous cases show both breakthroughs and limits
- These studies reveal how close apes can come to words

Impressive Linguistic Accomplishments of Nonhuman Apes

Example

Learned hundreds of signs or symbols

Understood spoken commands

Combined symbols creatively

Used gestures intentionally

Shared learned signals socially

Explanation

Apes like Washoe, Koko, and Kanzi acquired vocabularies of 100–400+ symbols, used appropriately in context.

Kanzi followed novel English sentences like “Put the soap in the water,” showing real comprehension.

Kanzi coined new phrases such as “slow lettuce” for kale, and “wet fruit” for watermelon, indicating flexible use of known terms.

Apes adjusted gestures based on audience attention—showing awareness of others’ perspectives.

Washoe and other chimps taught some signs to peers, hinting at rudimentary cultural transmission.

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Linguistic Limitations of Nonhuman Apes

The Limits of Ape Language Learning

- Apes share many foundations for language
- Primate simulation expanded representation and flexibility
- These same abilities may have evolved into symbolic thought

Example

Very short utterances

No consistent grammar or syntax

Mostly instrumental use

Limited generativity

Plateau in vocabulary size

Dependent on human training

Explanation

Most productions were one or two symbols long, rarely forming complex sentences.

Word order didn't affect meaning; combinations lacked structured relationships.

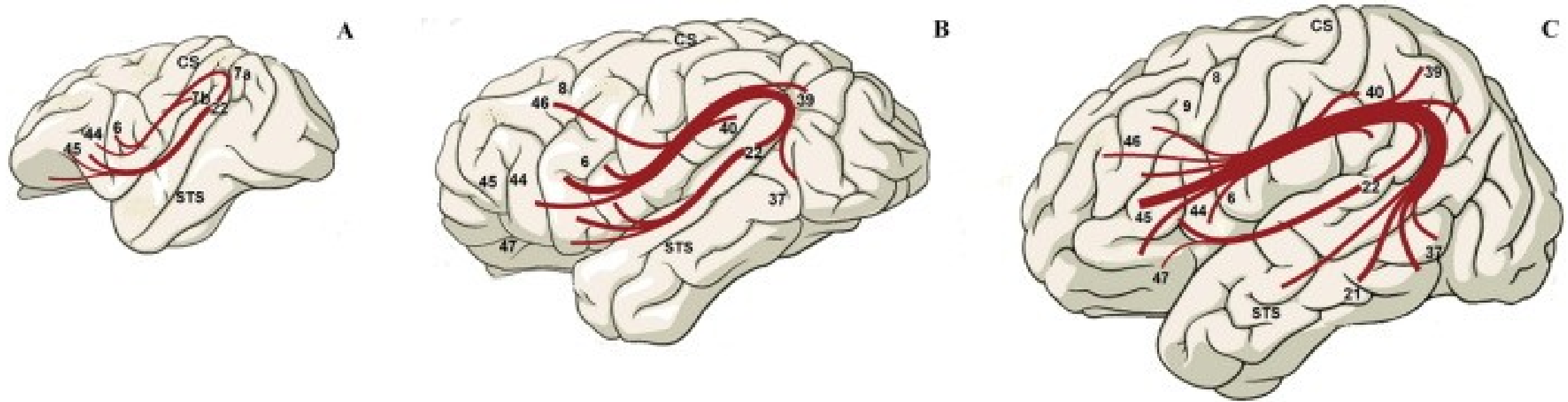
Communication focused on requests ("Give banana") rather than declarative sharing.

New utterances were rare—apes reused familiar phrases instead of creating new ones.

Progress slowed around the level of a human toddler; no explosion of new word learning.

Skills required intensive teaching and rarely emerged spontaneously in social groups.

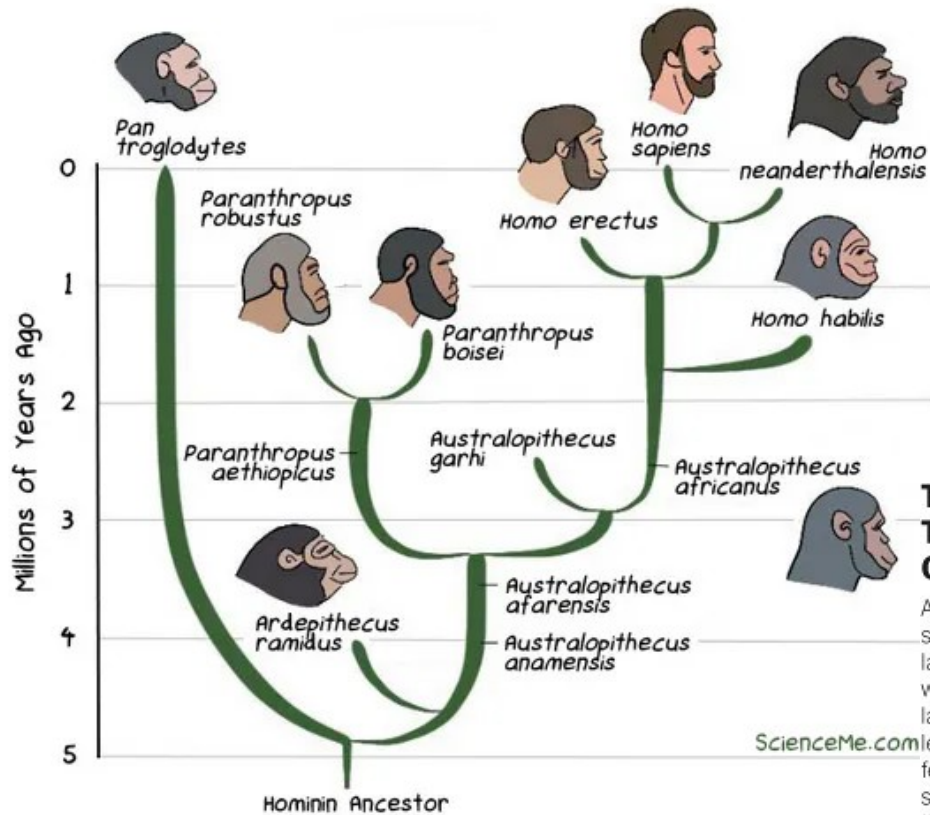
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From Simulation to Symbolic Thought

- Primate minds simulate self and others
- Flexible representations became symbolic variables
- These abilities paved the way for language

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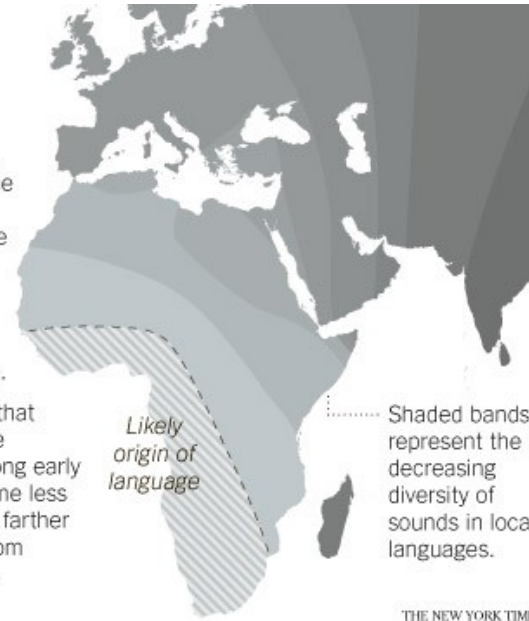


Tracing The Origins Of Language

An analysis of the sounds used in 504 languages around the world found that languages tend to be less diverse, with fewer phonetic sounds, the farther their speakers are from southwestern Africa.

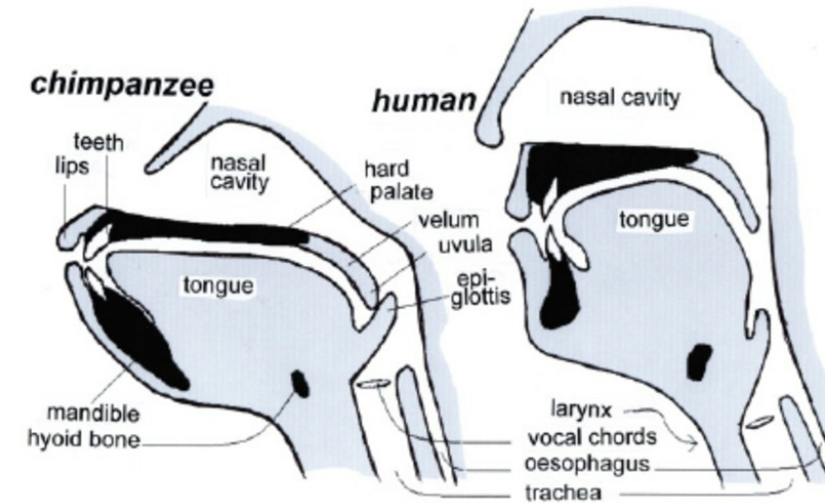
The pattern implies that language might have originated once among early humans, then become less and less diverse the farther humans migrated from southwestern Africa.

Source: Science

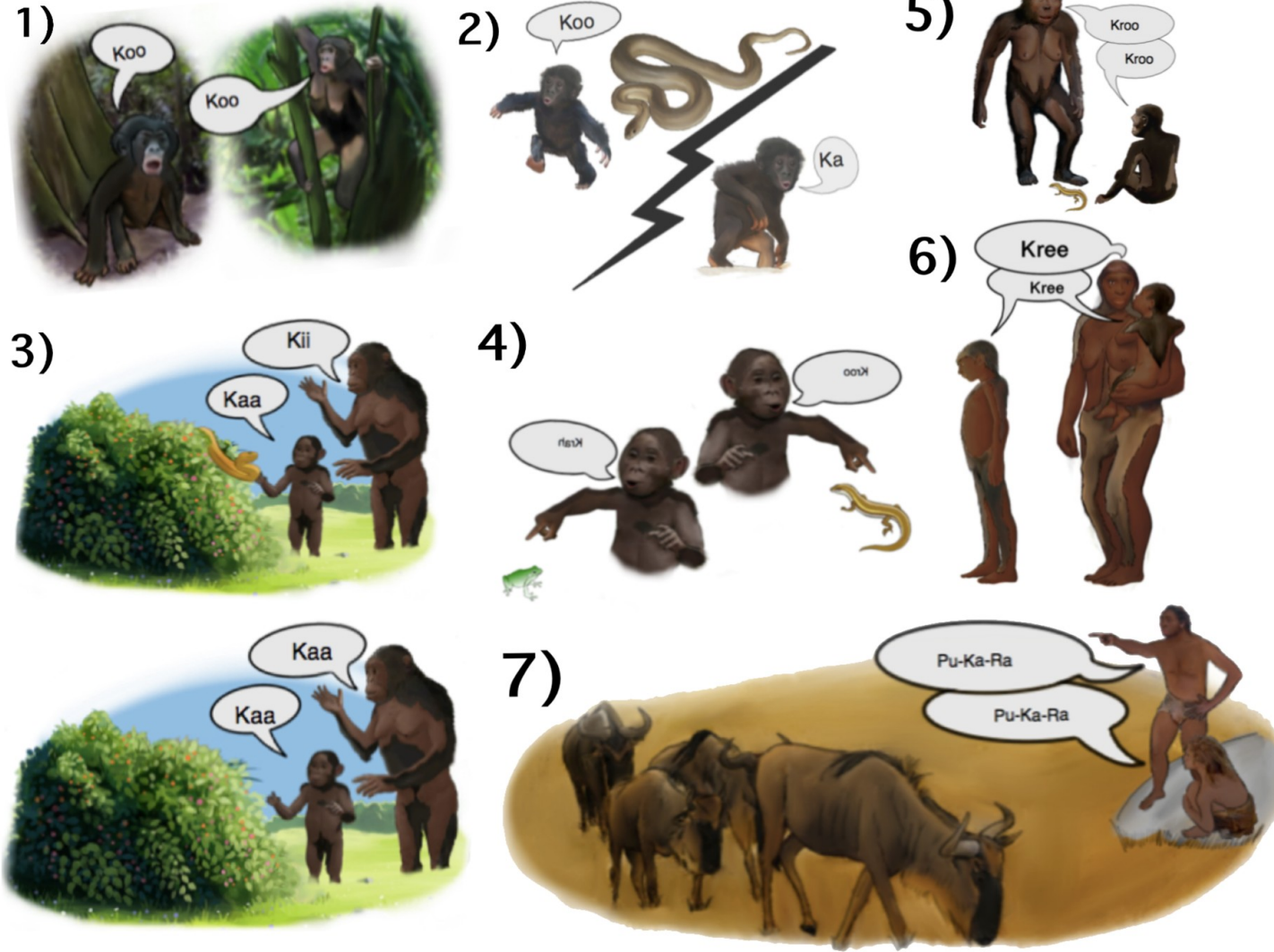


When and Where Did Language Emerge?

- Fossil and archaeological clues are indirect
- Where: Africa. When?
- Estimates range from early *Homo* to modern humans
- Language likely evolved gradually, not suddenly



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How Did Language Evolve?

- Language's origins remain debated and uncertain
- Different hypotheses emphasize different selective pressures
- The truth may combine several interacting causes

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		player X	
		cooperate	defect
player Y	cooperate	3, 3	0, 5
	defect	5, 0	1, 1

Why Did Language Evolve?

- Language is costly and complex
- It benefits others as much as oneself
- So why did it persist and spread?

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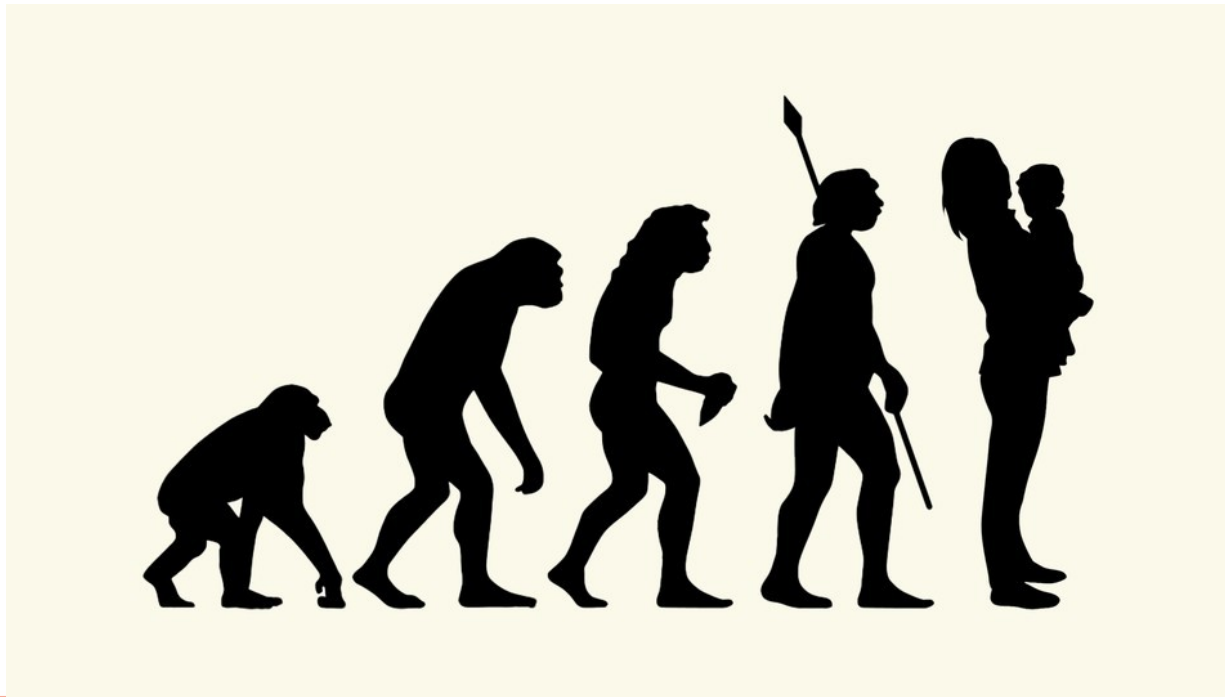
Adaptive Payoffs of Language

- Information sharing, teaching and learning
- Social cohesion
- Planning and coordination with others and oneself

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Language as a Social Lubricant

- Language first spread through close relationships
- Talking built trust, comfort, and attraction
- Speech became a social and emotional bond

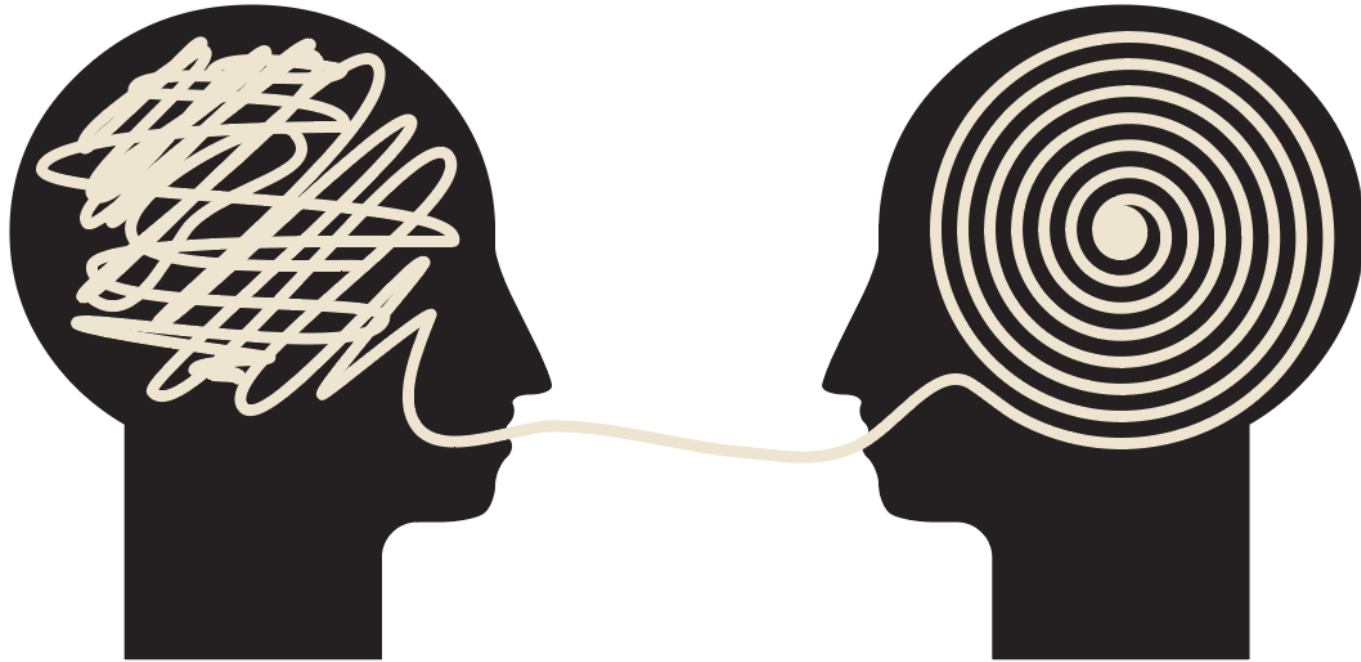


GROOMING,
GOSSIP, AND THE



EVOLUTION OF
LANGUAGE
ROBIN DUNBAR

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Language for Thought

- Language may have begun as inner speech
- Words help structure symbolic imagination
- Talking to ourselves made thinking more powerful

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	REINFORCING IN EARLY BILATERIANS	SIMULATING IN EARLY VERTEBRATES	MENTALIZING IN EARLY PRIMATES	SPEAKING IN EARLY HUMANS
SOURCE OF LEARNING	Learning from your own actual actions	Learning from your own imagined actions	Learning from others' actual actions	Learning from others' imagined actions
WHO LEARNING FROM?	Yourself	Yourself	Others	Others
ACTION KIND?	Actual actions	Imagined actions	Actual actions	Imagined actions

Cumulative Culture

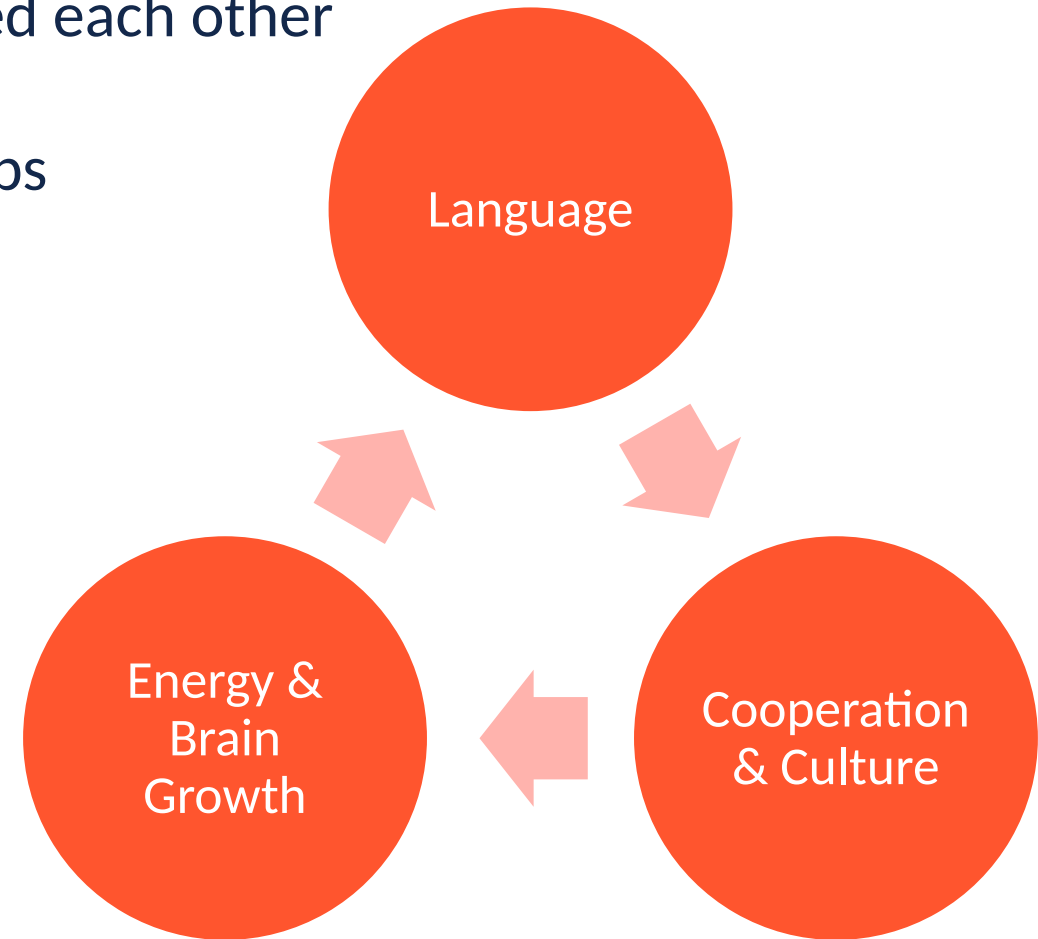
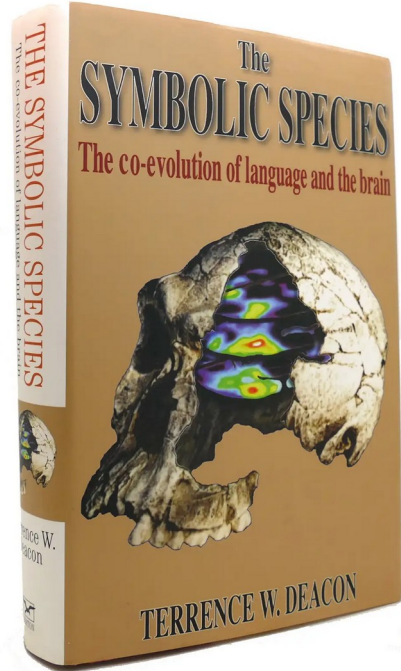
- Language made knowledge cumulative
- Ideas began evolving faster than genes
- Culture became a new kind of inheritance



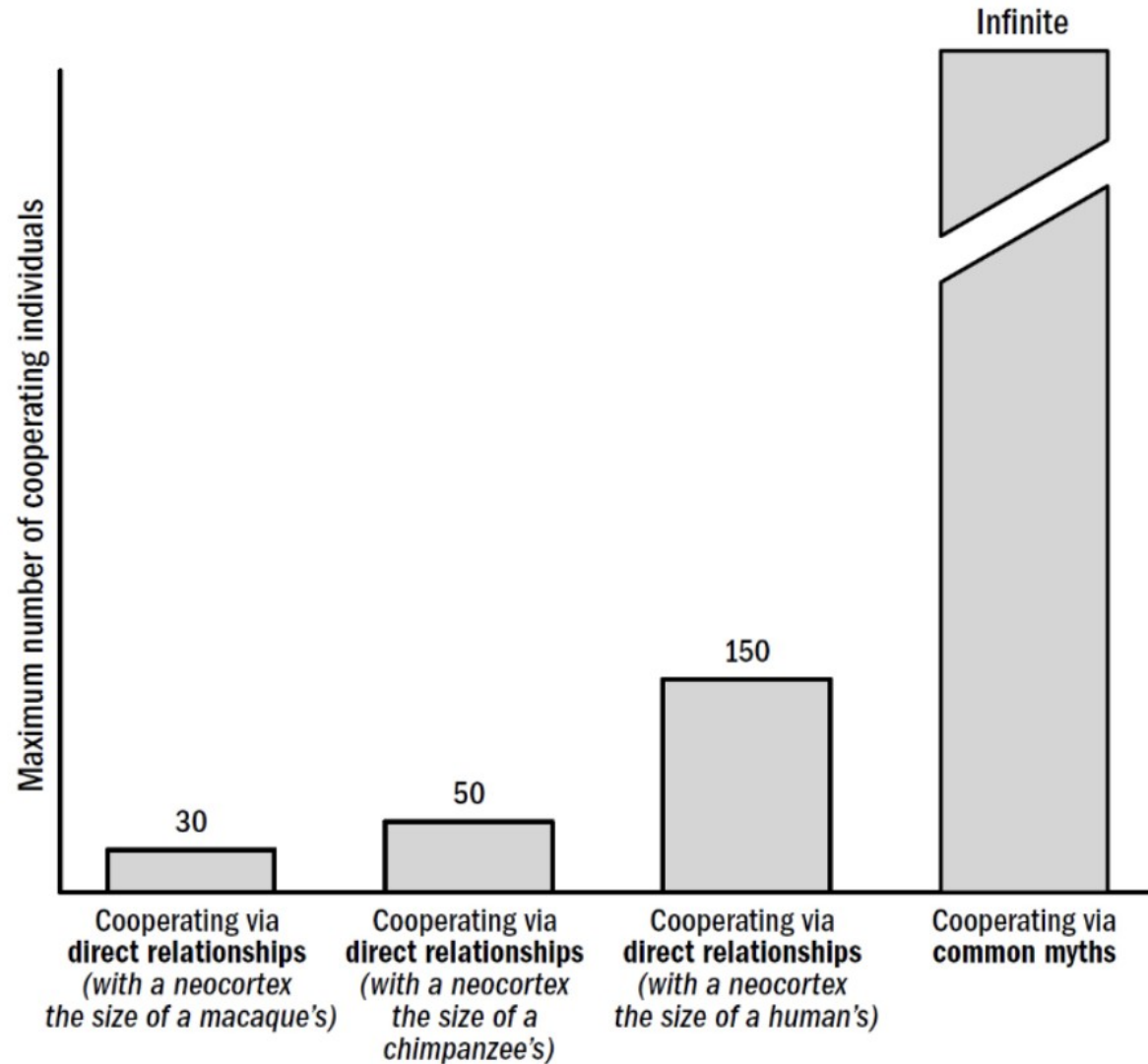
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The Runaway Feedback Loop

- Language, cooperation, and brain growth reinforced each other
- Bigger groups required better communication
- Better communication supported even larger groups



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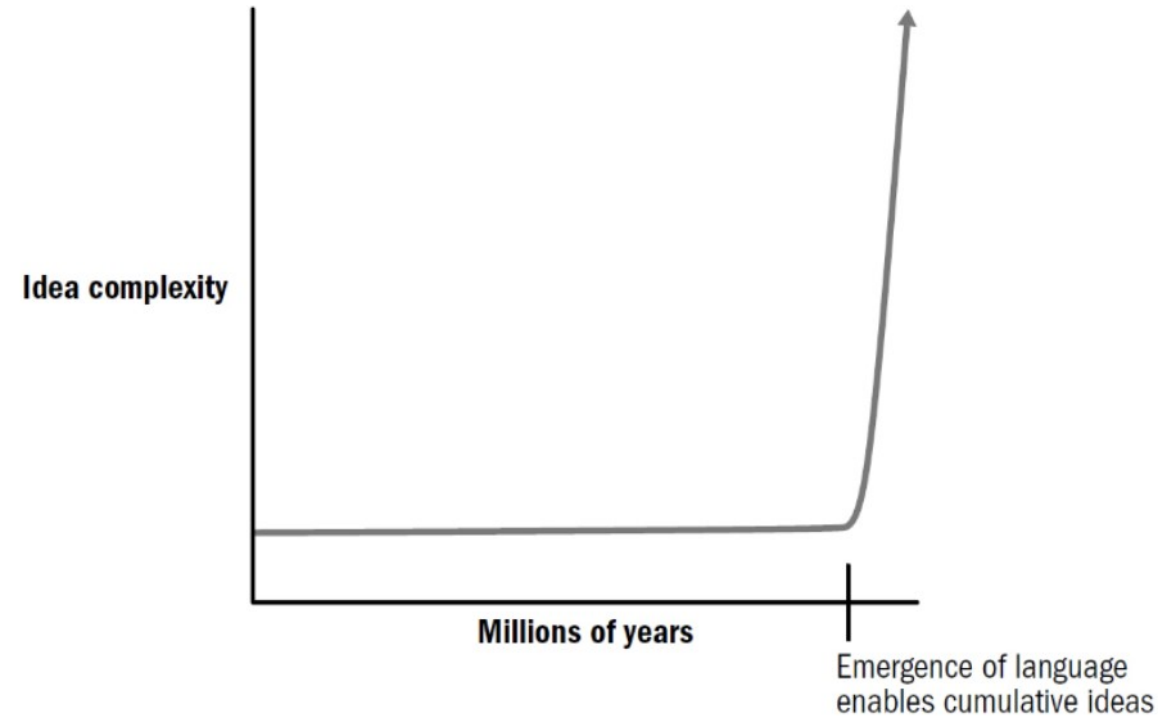
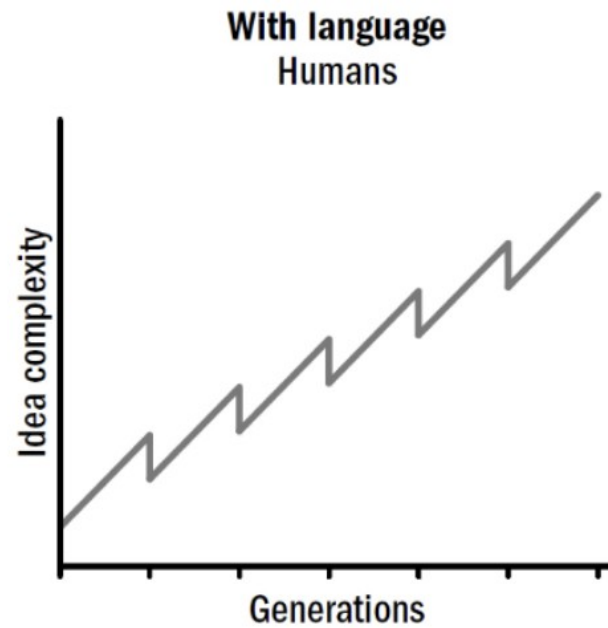
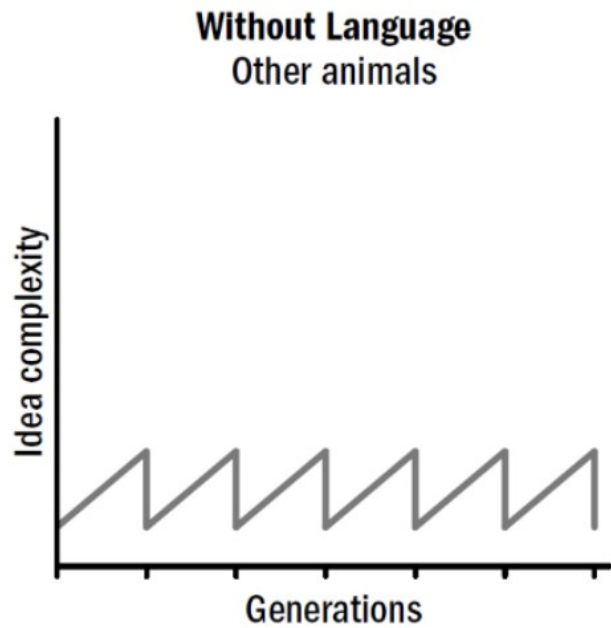
From Language to Symbolic Culture

- Language enabled shared imagination and abstract ideas
- Symbols allowed cooperation beyond direct experience
- Culture became a network of shared meanings

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The Singularity Has Already Happened

- The “singularity” usually means future machine intelligence
- Intelligence and idea starts growing exponentially
- But arguably it has already happened—with language and culture

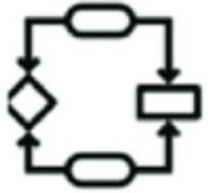


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Computation

What does the system do?



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Summary

- Language transformed communication and cooperation
- It built on primate social and cognitive foundations

Next Time

- How language actually works in the mind